

Ouroboros Code: A Practice-Based Study on Sci-Fi Atmosphere Design under Nonlinear Narrative Structures

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Figure 1: Screenshots of the four main scenes encountered in the game.

Abstract

This study investigates how atmospheric design enhances comprehension and emotional continuity in nonlinear sci-fi games. Using *Ouroboros Code*, a 3D interactive project, the researchers developed a design framework translating narrative and perceptual theories into practical methods for game environments. A mixed-method study combining theoretical analysis, observation, and user surveys identified three principles: atmosphere as emotional continuity, lighting as narrative rhythm, and spatial cues as cognitive anchors. Results show that atmosphere-centered visual systems improve narrative coherence and player orientation in fragmented story structures.

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1 Introduction

As interactive media evolves, game narratives have shifted from linear storytelling to fragmented, participatory systems. In science

fiction, time loops, parallel worlds, and memory fragments align naturally with nonlinear structures but also create cognitive challenges. Within such frameworks, environments must go beyond aesthetics to convey narrative logic, guide exploration, and evoke emotion. *Ouroboros Code*, an original 3D sci-fi game, explores how atmospheric design enhances clarity and immersion in nonlinear storytelling.

2 Our Approach

2.1 Methodology and Findings

This study employed a mixed-method approach integrating theoretical review, comparative analysis, observation, and survey to translate narrative and perceptual theories into actionable design guidelines for nonlinear science fiction environments.

- **Literature review:** The theoretical foundation was established through Böhme's aesthetics of atmosphere [Böhme and Calvé 2018], Murray's interactive narrative [Murray 2017], and Nitsche's spatial rhetoric [Nitsche 2008], which emphasize that atmosphere mediates emotion, spatial rhythm supports cognition, and player agency structures meaning. From this synthesis, three hypotheses were proposed: atmosphere sustains emotional continuity in fragmented storytelling; visual rhythm can substitute textual guidance; and spatial cues maintain cognitive orientation.
- **Comparative analysis:** To apply these hypotheses, a comparative analysis of three nonlinear titles—*Outer Wilds*, *Control*, and *Deathloop*—identified transferable strategies such as modular story distribution, dynamic lighting for pacing, and contrast-based landmarking [Jenkins 2004].

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- **User Study:** A user study was conducted to verify the hypotheses. Thirty-two participants explored two level prototypes developed in Unreal Engine 5. The experiment included three stages: an observation session recording player navigation and interaction; a survey assessing spatial clarity, emotional tone, temporal recognition, narrative coherence, and visual attention; and a short interview clarifying survey responses.
- **Questionnaires and Interviews:** Results showed that 76% of participants used color contrast to distinguish timelines, 81% recognized lighting changes as narrative turning points, and 69% relied on recurring symbols as spatial anchors. These findings support Lankoski's argument that perceptual cues enhance emotional continuity and player motivation [Lankoski 2010]. The empirical insights derived here inform the design framework summarized in Section 2.2.

2.2 Constructing the Nonlinear Narrative Structure

Theoretical and empirical findings informed a unified framework for nonlinear narrative design, translating abstract theory into three actionable principles:

- Distributing narrative fragments across spatial nodes.
- Sustaining emotional continuity through atmosphere.
- Embedding cognitive cues via visual rhythm.

Ouroboros Code applies these principles within a non-hierarchical system centered on memory reconstruction. Narrative fragments—including voice logs, visual residues, and symbolic artifacts—are distributed across four distinct environments, each defined by its own temporal logic and emotional tone. This structure enables players to reconstruct meaning through exploration rather than linear progression.

By converting theoretical insights into spatial design practice, the project demonstrates how narrative comprehension can emerge from environmental interaction, where atmosphere functions as an interpretive bridge, spatial cues guide progression, and perception becomes part of narrative assembly.

2.3 Narrative Functions of Atmospheric Environments

The theoretical framework is embodied in three key environments, each serving a distinct narrative function.

In the *Forest of Echoes*, glowing root systems encode fragments of an ancient language, encouraging players to infer lost memories through observation rather than explicit puzzles (see Figure 2a).

The *Altar of Remnants* features a central mirror pool symbolizing self-confrontation and identity rupture within the process of memory reconstruction (see Figure 2b).

The *Entropy Ruins* convey temporal instability and civilizational decay through floating debris and fractured geometry (see Figure 2c).

Each space is designed for visual readability and emotional resonance, enabling players to interpret meaning through perception rather than textual cues. Together, these environments translate

the abstract framework of nonlinear storytelling into concrete, experiential design.



Figure 2: The narrative function of environmental composition in scenes.(a)Forest of Echoes.(b)Altar of Remnants.(c)Entropy Ruins.

2.4 Visual Language and Cognitive Guidance

Building upon the environmental framework, the project defines a unified visual system informed by cognitive psychology and user experience research. This system integrates three interdependent dimensions—color, lighting, and spatial organization—each supporting player orientation and emotional rhythm.

- **Color System:** A desaturated blue-gray palette with selective high-contrast accents differentiates timelines and emotional states, drawing on established color–emotion associations where cool and warm tones convey contrasts such as reality versus memory.
- **Lighting Structure:** Dynamic multi-point lighting establishes rhythmic variations in tone and depth. Shifts in brightness and direction signal narrative climaxes and focus attention on key spatial or emotional elements.
- **Spatial Organization:** Layered, semi-open layouts and subtle non-Euclidean geometries represent temporal fragmentation, encouraging nonlinear exploration and reinterpretation of previously encountered scenes.

Operating on a principle of cognitive anchoring through visual recurrence, this system enables players to navigate complexity by recognizing evolving perceptual cues. Implemented in Unreal Engine 5 through modular level design and blueprint-driven triggers, it synchronizes environmental feedback with narrative progression, ensuring cohesion between atmosphere, rhythm, and story.

3 Conclusion

This study presents *Ouroboros Code* as a practical framework for translating narrative and perceptual theory into concrete design strategies for nonlinear game development. Integrating empirical research with creative experimentation, it demonstrates how atmosphere sustains emotional continuity, lighting structures narrative rhythm, and spatial cues provide cognitive anchors that enhance player orientation and engagement within fragmented story worlds.

By aligning theoretical analysis with design implementation, the research bridges academic inquiry and production methodology, offering a replicable approach for developers and scholars. The findings confirm that atmosphere-centered visual systems can preserve coherence and emotional resonance in nonlinear narratives, establishing a foundation for future work in interactive storytelling and immersive spatial design.

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